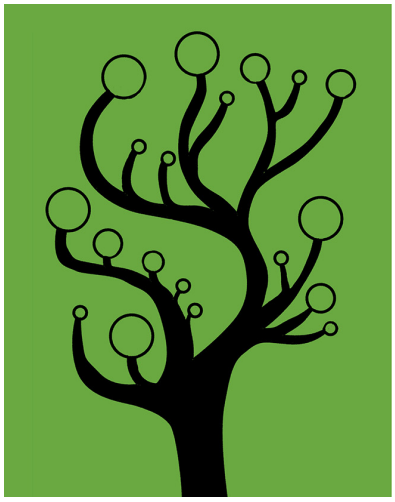


ORGANISMS CLEARLY CLUSTER INTO SEVERAL PRIMARY KINGDOMS

A MODERN VIEW OF DIVERSITY

IN CONTEXT

KEY FIGURE

Carl Woese (1928–2012)

BEFORE

1758 *Systema Naturae* (10th edition) by Carl Linnaeus classifies known life into two kingdoms: animals and plants.

1937 French biologist Edouard Chatton divides life into prokaryotes (bacteria) and eukaryotes (organisms with complex cells).

1966 German biologist Willi Hennig establishes a system of classification based on clades—groups of organisms based on common ancestry.

1969 American ecologist Robert Whittaker divides life into five kingdoms: bacteria, protists, fungi, plants, and animals.

AFTER

2017 A consensus among biologists accepts a seven-kingdom classification of life.

Before biologists had the equipment and techniques needed to scrutinize the microscopic structure of living things, biological diversity was split simply into animal-like and plant-like organisms. Then, in the 20th century, better